Chapter 4
Adopting a Funnel Strategy and Using Mind Mapping to Visualize the Research Design



4.1 Introduction

Academic research is more than a project or a piece of writing. A research paper, dissertation or thesis can only be written up after relevant research has been conducted. Research students will have different motivations for conducting a research inquiry, some will have a specific problem they would like to address through experimentation, while others will wish to add to theory. Designing a research study is often a major challenge for novice researchers due to a limited understanding of the research process and possibly also the particular meaning of research (Carlsson, 2006) in the context of a specific research problem. Many academic researchers are uncertain how to identify and craft the research problem to be addressed in the inquiry and how to frame the main research question. The beginner must conceptualize a research idea, set the scope of the research problem and consider whether the problem is researchable (Pajares, 2007). However, finding a researchable area may not be straightforward. This is where the funnel strategy is useful. The funnel strategy is a mental model, which a researcher can adapt, to provide guidance on the research process, enabling the flow of thoughts and ideas throughout the research journey.

4.2 The Concept of Research Strategy

A research strategy is the design of essential steps and activities that directs the view of the study and the resolve to conduct a methodologically sound research inquiry, in order to arrive at a valuable research outcome. It enables the researcher to remain motivated and focused, which can help lessen frustration, and improve the quality

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U. M. Mbanaso et al., *Research Techniques for Computer Science, Information Systems and Cybersecurity*, https://doi.org/10.1007/978-3-031-30031-8_4

of the inquiry. Time and resources can be saved by the adoption of an effective research strategy (Hauberg, 2011).

In the section that follows, we introduce the concept of a funnel approach to research strategy that enables the researcher to visualize the study process from the beginning to the end.

4.3 The Funnel Strategy Approach

The most important part of research design is formulating a researchable problem statement that clearly describes and establishes the scope of the study (Hassani, 2017). To do this, the researcher needs to create a logical process flow that commences with the identification of the broad subject area (e.g. cybersecurity (CY)), identifying the overarching research problem (e.g. cyber vulnerability) and three or four elements of the problem (e.g. cyber vulnerability challenges, practices and future needs). The design of the research study must then follow a rigorous processes of identifying the purpose or goals of the study, framing appropriate questions to interrogate the problem and devising ways to measure (using certain metrics for quantitative studies) or understand (using qualitative dimensions for qualitative studies) the phenomenon of study (Salhin et al., 2016). The research design includes choosing the data collection and data analysis method(s); planning, designing and conducting the investigations (through surveys, or interviews, or document review or a combination of these); and ensuring that the methodology enables the researcher to draw valid conclusions (Indulska & Recker, 2010). Furthermore, for the research outcome to be useful, or add new ideas to the body of knowledge, the research enquiry should focus on addressing everyday research problem(s) relevant to the economic and social challenges faced in a particular context or sector.

Ellis and Levy (2009) describe the concept of the funnel in a research study from the perspective of 'research-worthy problem (P)' described as the input to choosing a research area; 'the valid peer-reviewed literature (L)' as 'key funnel (components) that limits the range of applicable research approaches, based on the body of knowledge'; and 'the data (D)' that is relevant to the enquiry, which aids in sifting through the huge available data to find the exact study area.

The funnel strategy described in Figure 4.1 approached the concept from a different outlook, but with a similar thought. This funnel framework describes an initial approach to a research study and its design. Metaphorically, a funnel can be used to guide the thinking of a researcher on where to begin the research process, starting from a wider view of the subject matter under investigation. If we consider the wide mouth of the funnel as the broad field of study, for example, cybersecurity, and the narrow stem of the funnel as the specific research problem, then the research questions and research strategy can be derived from that, to guide the research.

Thus, the funnel framework can help to direct the researcher's thought process about how a research study can be carried out meaningfully in the real world. This is the psychological context of research, where the researcher engages in creating an

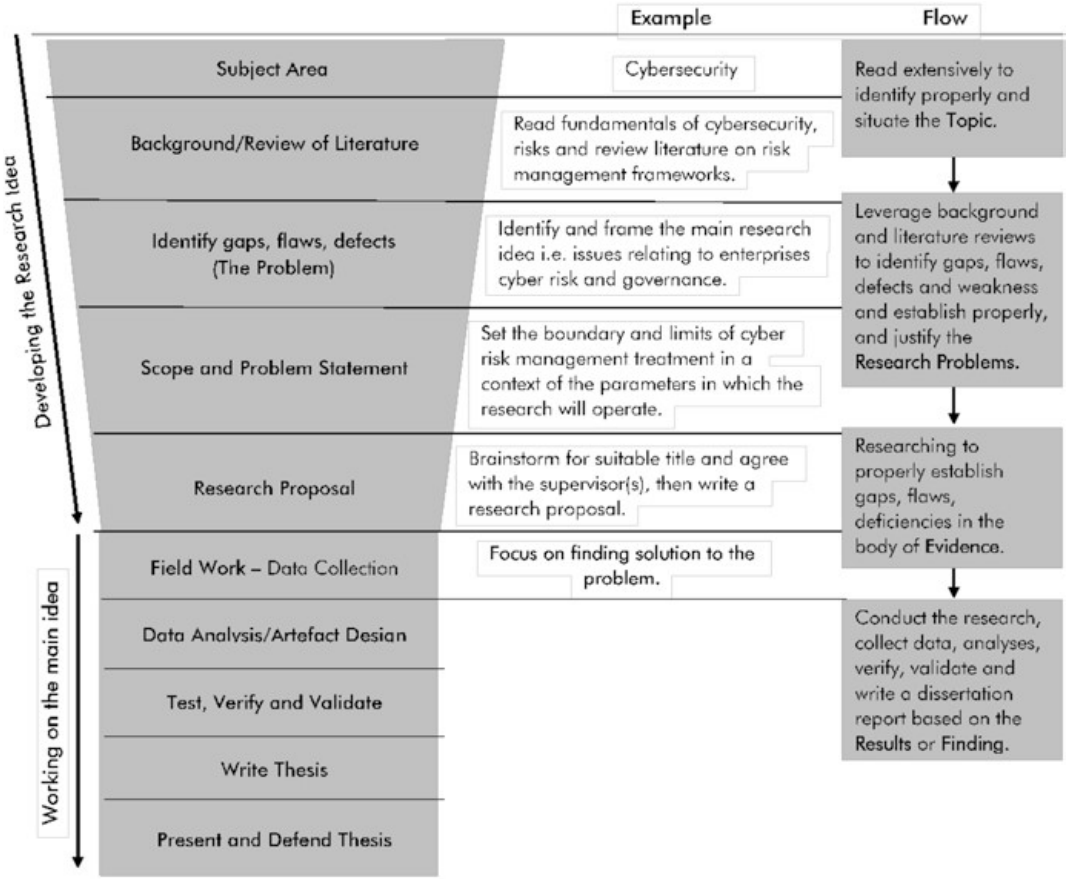


Figure 4.1 *The Funnel Strategy*

understanding of the relationships between the various research phases to direct her/his intuitive perception. The researcher needs to develop a sound, rational decision-making process, which is logical and evidence-based (MacIntyre et al., 2014). Clearly, a decision-making process can be described as a cognitive progression, capable of directing a course of action in a reasoned sequence, of which the outcome can either be convincing or illogical, based on particular assumptions or presumptions. Simply put, the funnel strategy builds the reasoning skills that are required in a typical research environment, including sense-making, constructivism and other thinking skills.

Sense-making is a way of creating situational awareness and thoughtfulness to appreciate situations of high intricacy or uncertainty and still make informed research decisions. Constructivism means that the researcher actively builds knowledge by constructing a view of the phenomenon based on data gathered from real people, who each have their particular interpretation of the phenomenon being studied (Ernest, 1996). Within the mass of data collected, the researcher grapples to build constructs, combining perspectives on the internal and external dynamics affecting the research problem. The funnel strategy can support this sense-making process, encouraging the researcher to filter the ideas encountered in the literature

while constructing the problem under investigation. In the sections that follow, the components of the framework are explained.

4.3.1 The Subject Area

A first step in the framework is identifying the broad subject matter of interest, in which the researcher may have some basic or little knowledge, for instance, the subject of cybersecurity vulnerability. Subsequently, having identified the broad subject field, the researcher must sieve through available scholarly literature, and other publications relevant to the topic, in order to ascertain that the subject matter is an area of interest worth devoting rigorous and time-consuming effort to pursue. This initial effort will require establishing a clear understanding with respect to intellectually stimulating knowledge gaps, or flaws in existing ideas or solutions, through a process of detailed note-taking. The researcher has to understand the ideas trending on the selected topic and must devise how the ideas relate to a set of broader or narrower associated concepts. More so, the researcher should create a list of keywords and phrases that can clearly express the topic under consideration. These keywords can help in advancing knowledge on the topic, which aids in expanding your knowledge foundations.

Building a knowledge tree: This phase of research is a critical stage that lays the foundation for the rest of the study. It serves a dual purpose of (i) helping the researcher to firmly grasp a broader view of the topic area and specifically (ii) ensuring that the focus/scope of the study has not been covered by other researchers or is not part of an ongoing research effort. This initial phase can uncover if there is sufficient information available to aid the investigation, very important for postgraduate researchers. The researcher or student should use this time to build focus, in order to avoid changing the topic. Each topic has its particular challenges, and proper scoping of the work will ensure that consistent progress is made.

4.3.2 Preparing and Writing the Background Discussion

The next step for the researcher is to gain considerable theoretical background and establish the underlying theories and concepts. The researcher should view the theoretical background as the catalyst for understanding, analysing and connecting the factors or variables that influence the study. Background reading enables the researcher to situate the enquiry, to understand the underlying theoretical underpinnings, essentially to ground the theories and concepts that can support the investigation of the phenomenon of study. Theories (e.g. socio-technical theory and theories of regulation) and concepts (e.g. digital innovation and cybersecurity regulation) are vital to explain the influencing factors and establish the domain of your particular research problem, relative to existing knowledge. Furthermore, the researcher must

engage with the real-world background data relevant to the problem, if the study includes a practice orientation, for example, analysing the development of broadband infrastructure and services in a selected African country or analysing the digital skills needed for effective cybersecurity operations. Without extensive background reading, the researcher will struggle to explain why the problem exists or to provide motivation for the appropriateness of the enquiry.

Consequently, the background section attempts to offer an understanding of theories and concepts that have relevance to the study and prepares the researcher to scope the problem space and domain. So the goal of the theoretical background is to reinforce the research in a manner that assumptions made can be critically evaluated to ascertain their appropriateness for the study. It serves as a link to existing knowledge, which fosters a better understanding of the research problem (qualitative research) or hypothesis (quantitative research) and propels the choice of research methods (Hassani, 2017). Besides, the theoretical background underscores the generalization of the various elements of the study and phenomenon under investigation. Theoretical backgrounds complement the review of literature in the area, noting that the guiding theories and concepts should be 'planted' in the background discussion and then discussed extensively in the literature review, where the researcher will identify the constraints and limitations of the existing theories and concepts. More so, focusing on specific variables or factors, this initial theoretical background aids the researcher to identify and locate the primary and secondary sources of data relevant to the research. Given the extensive information available online, keywords drawn from the initial reading material provided for lectures can be used to identify relevant sources of background knowledge.

Based on the funnel strategy, narrowing the focus to a specific topic of interest, such as *cybersecurity challenges experienced in a particular economic sector* (banking or medical information systems (IS)) in a particular country (e.g. Ghana, Kenya, Nigeria), the researcher should establish sufficient knowledge of the applicable theories and concepts in computer security, information security, network security, application security, people security, etc. The researcher then narrows the scope further to the factors and variables relevant to *cybersecurity challenges*, in the context of the topic. Subsequently, the researcher may highlight *cyber risk governance, cybersecurity professional skills* and *cybersecurity applications* in the banking sector as the three elements or variables to be examined in relation to cybersecurity challenges. These even more specific concepts can help identify the appropriate literature. In practice, the researcher begins to dig deeper into the particular topic, writing up detailed notes, in this case on *cyber risk governance* in the banking sector. From these notes, the researcher should be able to write a 1000-word background section highlighting the main research problem. Another example of the design of a research problem would be *identity theft in the banking sector*, stating three elements of this problem, for example, (i) types of threats due to possible abuse of personal information, (ii) financial losses and building customer trust and (iii) effectiveness of legislation and regulation in addressing threats arising from identity theft. This is the process of limiting the scope of the study to a particular clearly defined research problem and its specific elements.

One more vitally important point: At this stage, it is already necessary to use academic referencing style effectively, to cite all the works and sources being used. While there are a number of reference styles, and a particular course or university will advise which referencing style is required, we refer you to the APA seventh edition style guide, available at https://apastyle.apa.org/. Here you will find popular style guidelines, including title page setup, how to format your paper, bias-free language, in-text citations and reference examples. There are also webinars and tutorials freely available at https://apastyle.apa.org/instructional-aids/tutorials-webinars, for example, the webinar 'A step-by-step guide for APA student papers'.

4.3.3 Review of Scholarly Literature

A literature review is an important foundational step, which can help to build your knowledge foundations sufficiently to understand the problem domain, thereby arming the researcher with the ability to frame the research question clearly and concisely. Symbolically, this characterizes a narrowing step in the funnel stem. Thus, a researcher surveys scholarly papers including books, journals and conference proceedings relating to the subject area, compiles a descriptive summary of important ideas and themes and provides a critical review of the relevant works.

It is the expectation of the supervisor, the university, the principal investigator and fellow researchers that a researcher should spend ample time reading current journal articles on the subject area, as a way of igniting or directing the thinking and sense-making process. Figure 4.2 illustrates this initial phase of the research

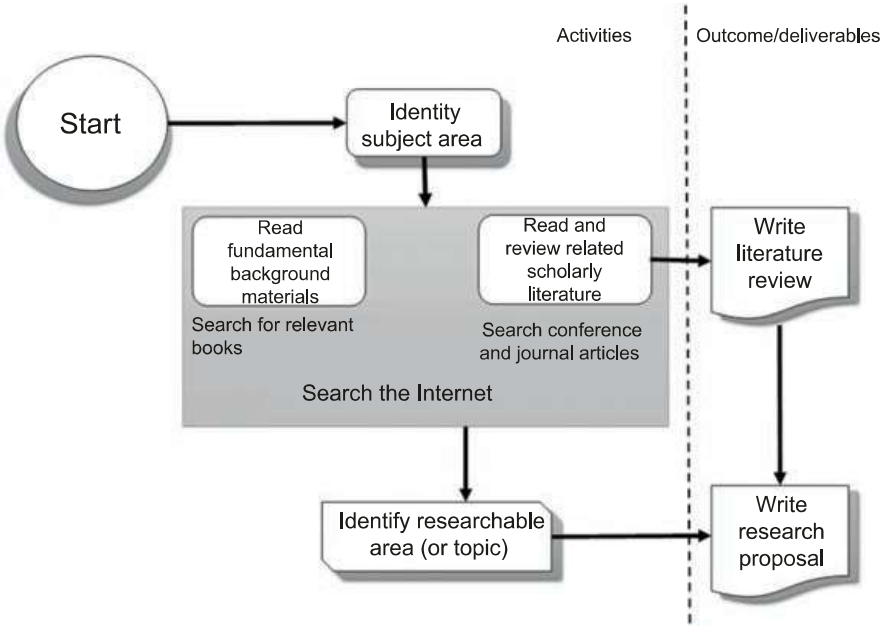


Figure 4.2 *Initial Phase of the Research Process*

process, demonstrating the steps that a researcher needs to undertake and the anticipated outcomes. As can be construed from the diagram, these are key activities that the researcher is expected to carry out within the wide mouth of the funnel, narrowing to the stem, in a logical sequence of activities, until the researcher has constructed a researchable topic.

Some researchers have argued that finding a researchable topic is not trivial, but a thorough literature review is capable of impelling the path towards a researchable topic (Pajares, 2007). This highlights the importance of identifying the main topic and associated keywords for the research, based on knowledge already gained from the background component of the study. The thorough literature review will help a researcher to understand the larger context of the study and knowledge already established about the topic. Conducting an extensive review of the literature in specific fields and exploring the references cited in those articles, book chapters and conference papers are an excellent starting point to your own research design.

At this stage, it is vital to think about the originality of your work and the possible contribution to knowledge. It is important to ensure that substantive work has not been done on the identical topic or alternatively that there is sufficient rationale to carry out further studies on the topic. It is also central to this part of the scoping process to ensure that there are sufficient sources of information to help the investigator substantiate the topic area. The literature review is discussed extensively in this chapter.

4.3.4 Identifying Gaps, Flaws, Defects and Weaknesses

The thoroughness of the literature review is not intended only to identify strengths in relevant works in the field but to find gaps, flaws, defects and weaknesses in existing studies. For example, the GDI (global diffusion of the Internet) framework was published in 2001 (Wolcott et al., 2001), at a time when public Internet use was in its infancy and when broadband services and mobile applications were virtually nonexistent. This requires a redesign of the framework, adapting the dimensions and levels to the digital environment of the 2020s, conducted by Mosina (2020). Identifying gaps, flaws and weaknesses demonstrates that the researcher has explored the existing knowledge to show where and how the proposed research fits into the field of study. It is expected that the researcher should substantiate how the literature contributes to the understanding of the problem domain under enquiry, by pointing to the strengths and gaps in the literature. The insights gained from the literature review also help the researcher to deal with conflicting concepts, resolve confusing issues and prevent duplication of effort, while justifying the necessity for further study. A detailed critical, analytical summation of relevant literature enables the researcher to scope the study and frame the research problem statement.

4.3.5 More Thoughts on Scoping the Research

The notion of scope brings the researcher to a vital decision point in a research study. It is concerned with establishing the boundaries of the research (what is included, what is excluded), so that the researcher can address the hypothesis or the main research question within set parameters. The objective is to clearly understand and precisely describe the limits of the study. Following the example in Figure 4.1, where the subject area is cybersecurity, it is noted that the field of cybersecurity is so vast that there is a need to narrow the topic to, say, cyber risk management. In this case, the researcher would focus on risk-related concepts and frameworks, not on cybersecurity in general. Thus, only background information, literature, theories and concepts relevant to cyber risk management will be included in the research problem statement, in the theoretical background and in the literature review. Thus, scoping is the product of the initial review of the existing body of knowledge in the general area of the field of study, that is, cybersecurity. The result of this scoping exercise is written up in the research problem statement, which states the main research problem and the three or four elements of that problem to be studied.

4.3.5.1 The Research Problem Statement

In academic research, writing a problem statement serves to improve the understanding and contextualization of the problem space and its significance. A well-understood problem is vital to resolving the problem and provides an effective solution (Pajares, 2007; Riggins & Wamba, 2015). In writing the statement, the research problem should be put in context, concisely, relevant to the problem domain, in such a way that offers the prospect of finding an answer, or practice-oriented response or solution or contribution to theoretical knowledge. It is important to understand that we need theory in order to build lasting solutions and innovations in any field.

4.3.5.2 Sample of Research Problem Statement

Over the past decade, financial institutions have sought to understand the factors contributing to large-scale cyber-attacks. The identity theft trend in the financial industry suggests that insiders' abuse of personally identifiable information (PII) accounts for many security breaches leading to identity theft and compromise of personal information. Identity theft accounts for huge financial losses in the banking industry which consequentially erodes customer trust. It has been argued that lack of effective cybersecurity governance, efficient legislation and regulation contributes significantly to identity theft in the financial services industry.

The above statement suggests that a quantitative or qualitative approach can help determine how the various factors of governance, legislation and regulation can

affect identity theft. However, in typical quantitative research, the hypothesis can provide a statement to be tested using quantitative methods.

4.4 Design Project Versus Research Project

Not every project is an evidence-based research study. It is important to understand the difference between a design project and a research project (Hevner et al., 2004; McLaren & Buijs, 2011). The difference between the two is the underlying approach and process. In the digital sphere, one obvious characteristic of a design project is the adoption of an already established software development life cycle (SDLC) approach, without any attempt to refine the methodology. On the other hand, the research project will have a philosophical underpinning and apply investigative methodologies to probe the unknowns of a particular phenomenon (Hevner et al., 2004; McLaren & Buijs, 2011). Furthermore, a research study goes through the processes of data collection, data analysis, argumentation, explanation, verification and validation and may include implementation and testing (Oates, 2006). Thus, the research project is concerned with establishing new knowledge, whereas the design project is purely concerned with solving a particular problem without regard to new knowledge. While the research project is an academic work, which aims to contribute to the body of knowledge, in contrast, the design project seeks to solve a problem purely with existing knowledge. However, the outcome of academic research work may include the production of digital products or artefacts; thus CS (computer science), IS and CY research can include design creation methodologies (Oates, 2006).

The point here is that in some instances, research in CS, IS and CY is not satisfied with only the theoretical findings of the research study but requires the construction of a formal proof of concept, or prototype, to demonstrate the applicability of empirical findings, by using an established design and sound engineering principles. The idea is for the researcher to create a valid, repeatable scientific process to demonstrate that the outcome of research provides a solution to the real-world problem. In other words, a research study in CS, IS and CY can apply design principles to create proof of concept, based on the initial research findings.

For any computer science-based research output, the researcher is expected to demonstrate that the work is not a standard design or copy. The researcher must demonstrate those academic skills relating to applicability of methodologies, analysis, explanation, justification, argumentation and critical appraisal, alongside the attendant creation of new knowledge, not just demonstrating technical skills. An example of this is using SEMAT Essence Kernel and IBM Garage methodologies to analyse the strengths and weaknesses of the design process for an application for pest surveillance in agriculture, in order to improve the future app design (for further information on SEMAT [Software Engineering Method and Theory] see www.semat.org). Novelty and risk-taking are the hallmarks of research endeavours. Thus, researchers and students must distinguish the research project from the industry-oriented design project, noting that this should be done from the outset.

4.5 Developing the Flow of Ideas

The funnel strategy diagram depicts the mental process that generates the flow of ideas that should be followed in the design of a research study, without setting out the details of the tasks themselves. Many tasks in research work will require the researcher to develop a flow diagram, guiding the research process. Figure 4.3 illustrates a typical flow of ideas to develop a research plan and includes the activities that are required at each step.

The first step is to narrow down the range of possibilities to the particular topic of interest. The researcher must understand the motivation and state it clearly and concisely. With this in mind, the researcher seeks to develop a statement that describes the particular research problem and the reason why addressing that problem is imperative. Confirming that the problem exists requires a concerted effort, through the review of relevant background information available in reports and other documents, as well as the most relevant scholarly literature in the field. By navigating from one set of documents to another, and from one set of thematic articles to another, the researcher can identify and discuss the related arguments or positions, thus establishing the evidence for the nature and characteristics of the research problem. Weaknesses, defects, flaws, shortcomings and suggestions for future works in those articles can guide the researcher to the focus for your particular study.

The development of ideas can be stimulated by certain questions that the researcher poses to her/himself, such as the following:

- Where does the problem lie? Who does the problem affect? What is the context?
- What is the overarching research problem?
- What are the three key related elements of this research problem?
- How will it affect the economic sector, or social sector, if the findings are able to address the problem?
- Is there a timeline to fix the problem?

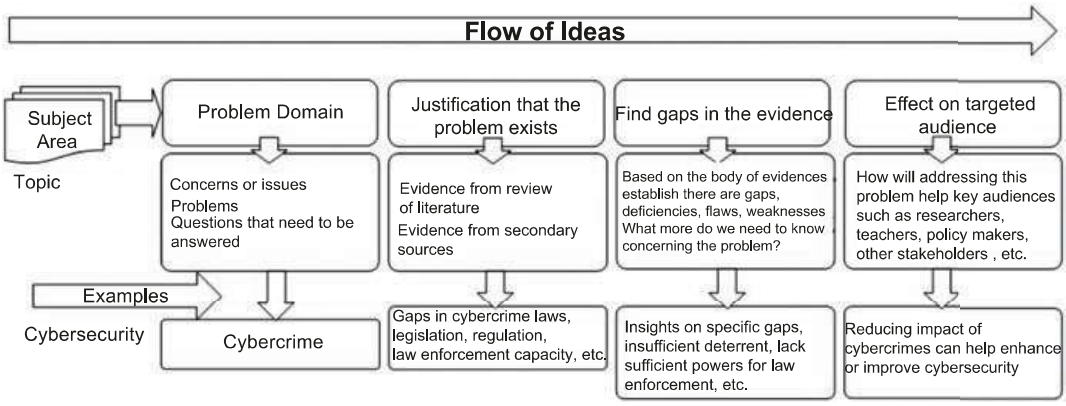


Figure 4.3 *Typical Research Task Flow*

- Is it all important to fix the problem?
- Is the problem a temporary issue or is there a likelihood that it continues into the future?
- Is the research capable of revising existing knowledge or practices? If so, how?

Once the answers to these questions are provided, the researcher should have a well-rounded problem statement and make a few drafts until the problem statement is as polished as possible.

Therefore, we can conclude that the funnel strategy stimulates the flow of ideas in the early phase of research design in a logical and structured manner, and using this approach can be a valuable asset to the researcher. In CS, IS and CY research, it is easy to be side-tracked by the sea of information that is accessible. By using the funnel strategy, the researcher can direct herself/himself to remain focused. This will allow the researcher to achieve better outcomes and avoid wasting time and effort due to taking unnecessary detours away from the core goal.

4.6 Mind Mapping Applied to Research Design

4.6.1 The Concept of Mind Mapping

Contemporary research requires a well-structured sequence of logical and structured thinking, and a strong flow of ideas, to drive an acceptable scientific or social enquiry. This is challenging to many new researchers. Crafting research topics, perspectives and problem statements, from the apparent chaos of ideas in the literature into logical, structured and creative arrangements, combined with a flow of logical thought patterns, is necessary. The researcher can use mind mapping to derive aims, design objectives and write up the hypothesis (quantitative research) or problem statement (qualitative research). Mapping ideas enables us to organize ideas into coherent thought (Wilson et al., 2016). It helps to increase the flow of understanding and makes fresh connections highly visible, to create many possibilities of arranging thoughts, ideas and relationships in a logical order. The technique fosters creativity by enabling intuitive thinking that builds on existing knowledge. It combines appropriate logical and structured thinking, with intuitive thinking. Researchers can derive hypothesis, problem statements, objectives, questions, approaches and methods and explore validation through mind mapping. Usually, visual organization of ideas is based on hierarchical order showing the appropriate relationship among related concepts (Crowe & Sheppard, 2016); however, mind mapping approaches and tools enable the researcher to draw many distinct kinds of relationships, criss-crossing themes and categories and showing dynamic linkages across a broad set of ideas. A sub-topic, concept or theme can be mapped onto the main topic, which allows integration of several ideas in a nonlinear, graphical layout that enables the building of an intuitive framework around a central concept (ThinkBuzan, 2010).

Mind maps are graphical representations of data, information and thoughts, as opposed to traditional linear notes or annotations in text documents. Consequently, mind mapping can aid the researcher to capture thoughts, ideas and keywords and organize them in a multidimensional structure. The central idea or title is always placed at the centre of the mind map, and related ideas branch off from the centre in multiple directions, thereby establishing a radiating structure. A mind map can be accomplished with manual drawing, using rough notes from meetings or technical sessions, and there are also a range of automated software tools that can aid in mind mapping (see, e.g., Software Testing Help at https://www.softwaretestinghelp.com/mind-map-software/). With mind mapping, it becomes easier to brainstorm and then organize thoughts into structure and order. It assists you to visually structure your ideas for synthesis, analysis and recall.

4.6.2 Mind Map Use Cases and Benefits

Mind mapping techniques can be applied to any research process, converting unstructured information into a memorable and organized structure. It enables the researcher to visualize the bigger picture and to build connections and hierarchies of ideas. It activates the brain and enhances the thinking process by applying mental triggers. In addition, mind mapping empowers unrestricted flow of thoughts and enhances memory. By using mostly single keywords, symbols and phrases in creating mind maps, the researcher can jot down thoughts in a faster way, which encourages an unhindered flow of ideas.

Mind maps are not meant only for applications in research but can be applied to many diverse tasks, including lecture note taking, as a study aid, brainstorming individually or in groups, problem-solving, gaining insight into complex subjects and research project planning. By so doing, research productivity is increased in a time-saving manner. Figure 4.4 shows a typical mind mapping exercise in the academic research context. It is very useful in writing research proposals since it provides researchers with a logical way of thinking, as well as building the timeframe for the research design.

4.6.3 Application of Mind Mapping to Setting Out the Research Tasks

Using mind mapping to draft the outline of your research proposal and plan the layout of your thesis can be a significant contributor to achieving success. The mind map guides the entire research process, allowing systematic flow in the research approach. When applied to setting out the tasks for the research proposal,

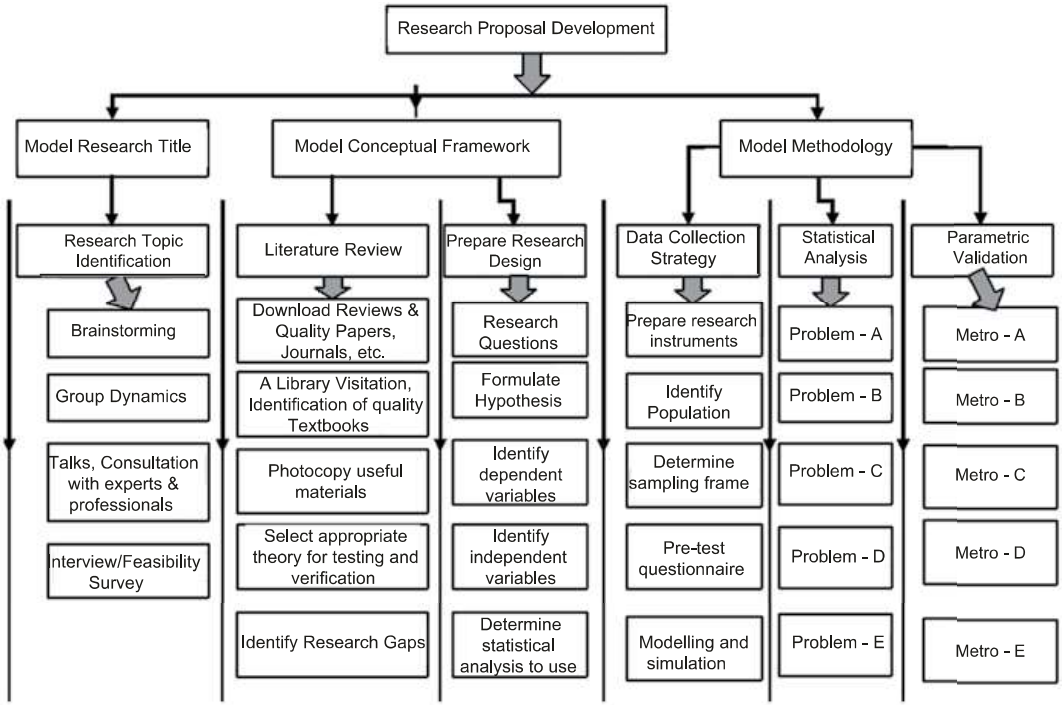


Figure 4.4 *Mind Mapping Framework for Research Models/Proposals*

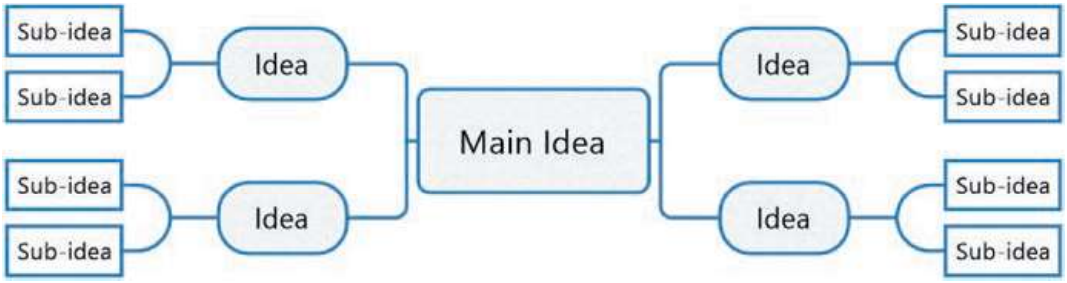


Figure 4.5 *Idea Synthesis in Mind Map*

it allows you to plan extensively, providing critical details that need to be addressed, within a set time frame, followed by monitoring the planned progress. The mind map illustrated in Figure 4.5 presents a mind map format for your ideas or research tasks.

In order to create a mind map for writing the research proposal, the researcher should consider the following steps:

Step 1: Identify the main goal.

Since the intention is to prepare the research proposal, the researcher's goal will be highlighted as *Prepare Research Proposal*: Type this central idea in the 'main idea' box.

Step 2: Identify the activities needed to prepare the research proposal.
Preparing a research proposal starts with three components:

1. Component 1: The selection of research title, problem and purpose statement research questions or hypotheses
2. Component 2: The building of a logical or reasonable conceptual/analytical framework
3. Component 3: Determination of methodology and methods

Step 3: Determine and specify the tasks that must be achieved.
Here you can set out the major activities and the corresponding action points:

Major activity 1: Preparing the research title
Action point: A suitable title must identify the research topic that will be the focus of the study, highlighting the dependent and independent variables (quantitative research) or the main problem and at least one of the elements of the research problem (qualitative research).
Major activity 2: Building the conceptual/analytical framework
Action point: The conceptual or theoretical analytical framework requires a detailed literature review, time for reading papers from scholarly journals, selecting appropriate concepts and theory(ies) to guide the research, identifying gaps in knowledge to be addressed, drawing an analytical framework diagram and including a short explanation of the diagram.
Major activity 3: Developing the research methodology
Action point: Based on the research gaps, objectives or questions identified, methodology design tasks include specifying the following:

Activity 3.1 Data collection approach: This involves preparing the research instruments, identifying the study population, determining the sampling frame and pretesting the questionnaire (quantitative survey) or the interview guide (qualitative study).
Activity 3.2 Statistical analysis: This requires proper understanding of statistical tools and how to carry oust statistical analysis for each hypothesis, where the study is quantitative. Parametric validations are also vital to validate the researcher's expected/proposed result.

These steps can be modified while assigning timeframes for each of the activities that you have identified. In so doing, the researcher will be able to make a rough estimate of how long it will take to be able to prepare the research proposal. If the researcher has a fixed period within which to submit the draft proposal, the researcher can allocate specific periods for the accomplishment of each activity, which can facilitate submitting the proposal on time. Figure 4.6 gives a complete picture of the mind map applied to writing the research proposal. Leveraging this mind mapping approach will produce a good quality research proposal, noting that you also need to pay attention to the quality of the content.

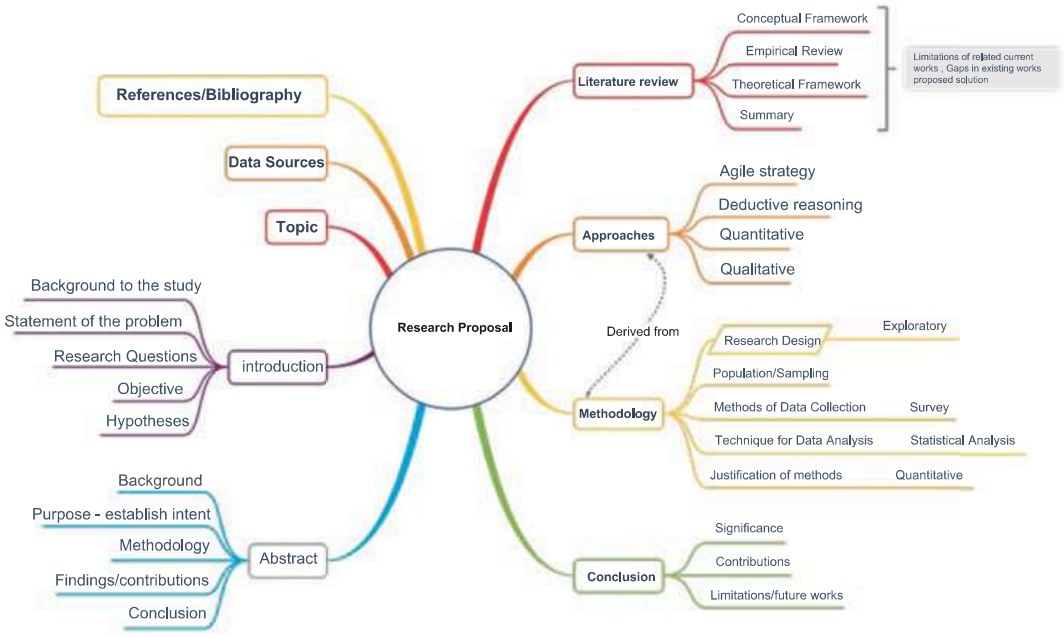


Figure 4.6 *Mind Map for Research Proposals*

4.7 Applying the Funnel Strategy and Mind Mapping to the Research Content

In this section, an attempt is made to demonstrate how the funnel strategy and mind mapping can stir a researcher in conceptualizing and conducting a research study. The thrust of these two tools is to direct and aid the flow of thoughts, which can drive the concept mapping of the research parameters and connects existing pieces of evidence to the fresh ideas developed. So we begin with the central idea of the subject area or topic and work through to situate and address the problem in the context of the research constraints. The following diagram explains how to apply the tools described in this chapter.

4.7.1 Example 4.7.1: Autonomous Agent and Multiagent Systems

Figure 4.7 illustrates how the funnel strategy is applied to conduct research in the area of autonomous agent and multiagent systems.

Having identified the broad topic of interest through learning about the field and conducting an initial scan of journal articles, the next step is to engage in extensive background reading of established concepts and theories relating to the topic. For instance, most complex intelligent software systems require a good knowledge of algorithms, theories and assumptions, so learning from existing ones, and gaining

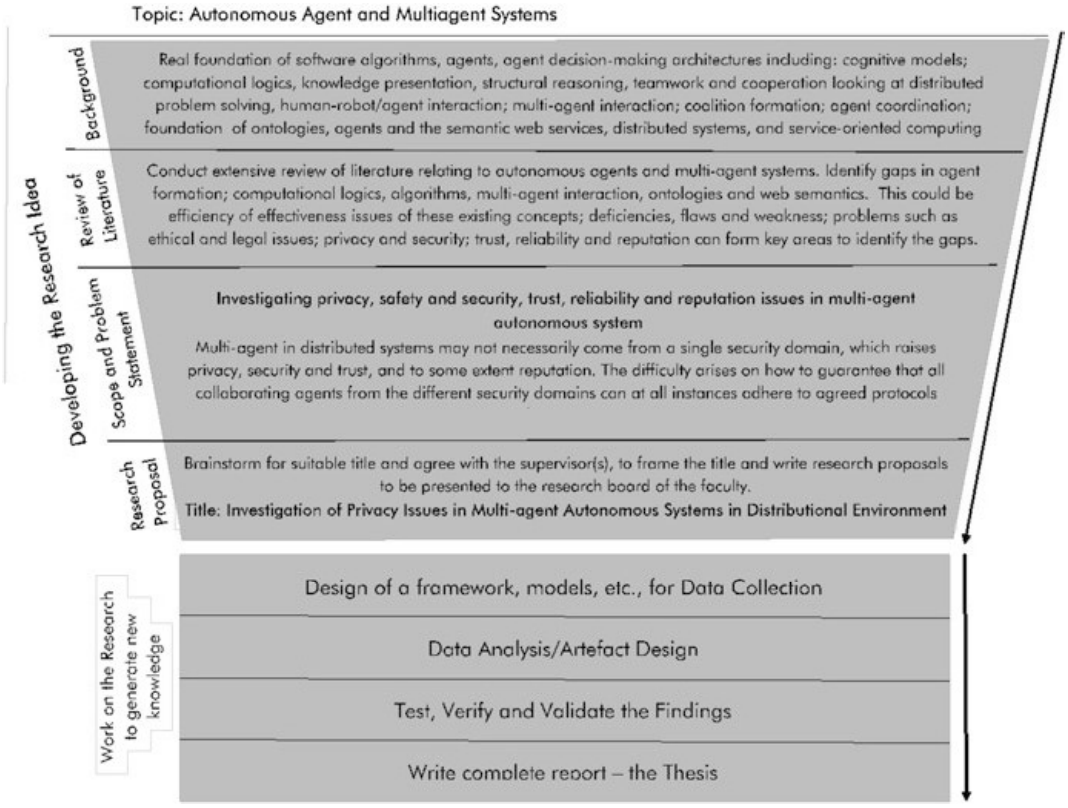


Figure 4.7 *Example Application of Funnel Strategy*

an understanding of how to create fresh algorithms should help build the researcher's intellectual capacity. Software agents are built based on sound decision-making, cogent reasoning, computational logic and communication protocols. It is important to engage in extensive background reading with respect to the key foundational concepts relating to the topic, to sharpen the researcher's knowledge and understanding, before reviewing the literature in depth. The knowledge gained during the extensive reading on foundational concepts will help the researcher to understand and present in writing these concepts found in literature, in your own words. Figure 4.8 depicts how to apply mind mapping as a way to do concept mapping on the research topic. Radiating out from the centre text box, the first-order elements randomly show important sub-topics, and subsequently, branches are created to reflect the lucid thinking of the researcher with respect to the concepts, important variables and parameters relevant to each sub-topic. The supposition is that if a researcher can carefully and systematically learn from previous research works, she/he can comfortably frame and construct a new research enquiry. When fresh thinking is interconnected with an existing body of knowledge, then that fresh thinking is a lot easier to present, explain and write up in the text of the research proposal. So the researcher has the responsibility to create a scaffolding of ideas from an existing body of knowledge, in order to introduce fresh knowledge. Doing concept mapping is one way to scaffold ideas and design research parameters (what is included in your study, what is excluded from your study), also referred to as defining the scope or boundaries of the study.

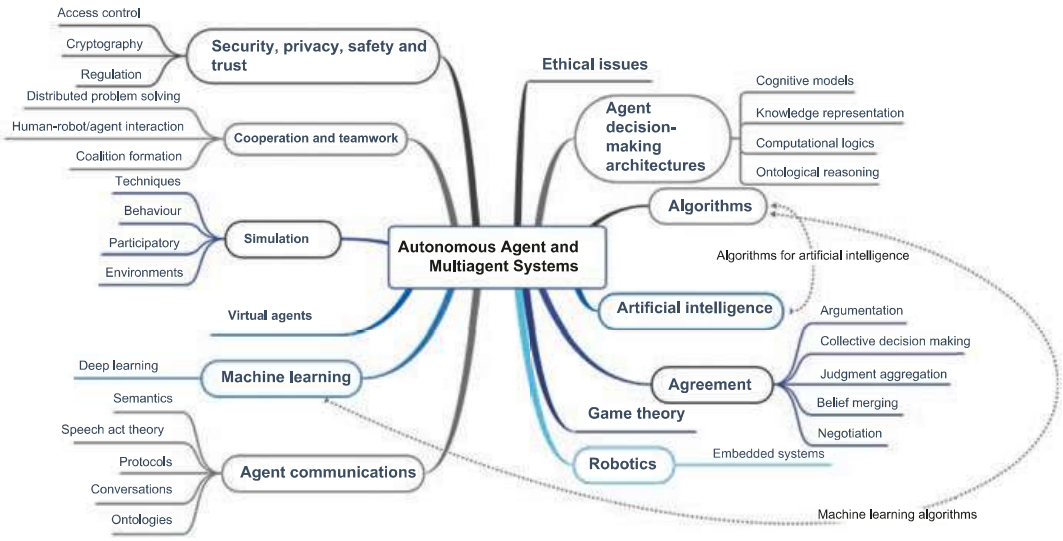


Figure 4.8 *Concept Mapping of Autonomous Agent and Multiagent Systems*

4.7.2 Example 4.7.2: Artificial Intelligence (AI) and Robotics

Similar to the example discussed in Section 4.6.1, a research enquiry in the subject area of artificial intelligence (AI) and robotics can cut across several disciplines, requiring the researcher to contextualize the investigation in a multidisciplinary or alternatively in an interdisciplinary way. Figure 4.9 depicts the topic of artificial intelligence and robotics, narrowing to the possible domains or sub-topics within this subject area.

What this entails is that the researcher will focus on the conceptual and theoretical frameworks that deepen the understanding of background information on any of the sub-topics. However, the researcher must gain a thorough understanding of the basic AI and robotics subject area, before diving deeper into the context of the sub-topics. The researcher must seek to understand the present 'state of the art' and imminent research needs and to identify important research gaps, through connecting the background information, concepts, theories and assumptions underlying the study, to the discussion of those concepts and theories in scholarly articles. The researcher must understand the foundational computer science concepts needed to enable her/him to conduct research in AI and robotics, notwithstanding which particular direction the research will take. Applying the funnel strategy, background reading is the first step, after the researcher has chosen the sub-topic(s).

Figure 4.10 demonstrates the fundamental constructs that can underpin the understanding of AI and robotics and further learning. For background reading, the researcher must comprehend such concepts as neural networks, machine learning, data mining, deep learning, big data, intelligent systems, algorithms, kinematic transformations, reverse kinematics, etc., regardless of the area of concentration.

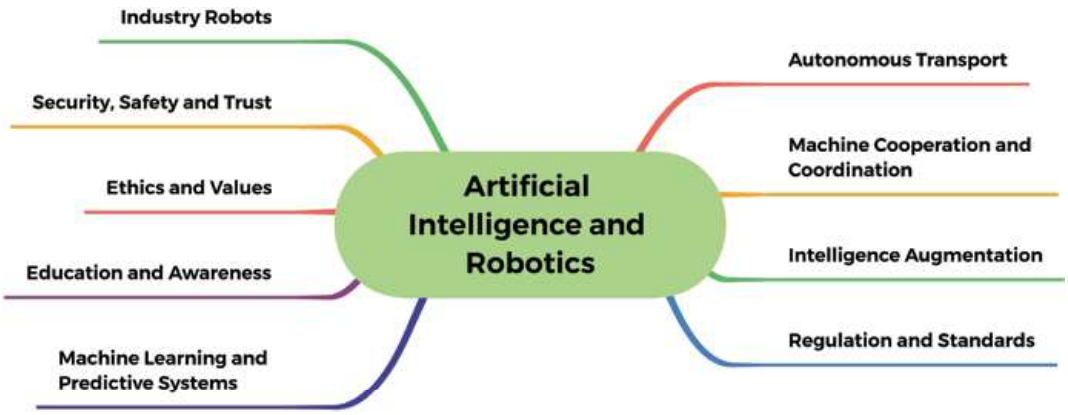


Figure 4.9 *Artificial Intelligence and Robotics Sub-topics*

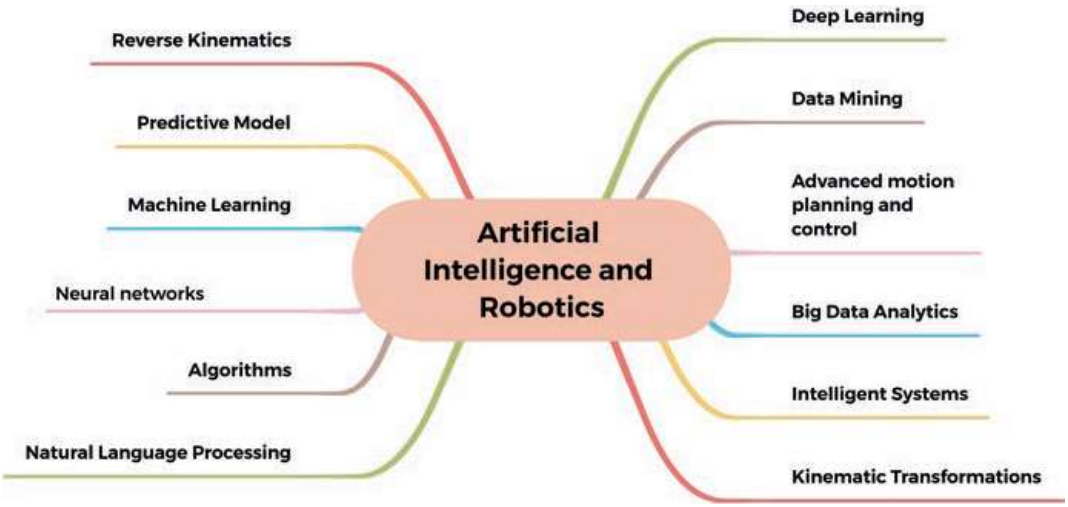


Figure 4.10 *Foundational Constructs in AI and Robotics*

Generating a clear understanding of these concepts in the proposal will assist in building a deeper understanding of the topic and sub-topics, creating the path to situating and scoping the study. Reading as widely as possible and understanding relevant theories and concepts is necessary to writing a good literature review, where the most important ideas are discussed, and the research gap is clearly identified. Having scanned and read widely, the next step is to select those journal articles and peer-reviewed conference papers, theses and dissertations that will be used for the literature review, all guided by the decision on the scoping of the topic and sub-topic. For example, the selected sub-topic could be ethics and values. In this case, the researcher can narrow the scope of the study to identifying the gaps in research in ethics and values. The next step is to brainstorm contextually and frame the



Figure 4.11 *Concept Mapping Extension of Ethics and Values*

problem to be addressed and decide on a suitable research title. For instance, such research title can be framed as *investigation of ethics and values in autonomous transport systems*. The next step is to develop the research proposal to reflect the chosen topic.

At this point, the researcher has arrived at the stem of the funnel, which metaphorically illustrates that the researcher has finally found the direction of the study, and henceforth can sustain and remain focused until the research is completed. In the case of this example, we now use a mind map to draw the underlying concepts and parameters to suit the specific scope of the study. Figure 4.11 illustrates a conceptual perspective of AI ethics and values that the researcher can apply to the specific context of autonomous transport systems.

The mind maps shown in Figure 4.5 through to Figure 4.11 show the cascade of ideas that is possible through the use of mind mapping tools and techniques. With this sort of mapping, the researcher is armed with an array of possible alternatives to narrow and scope the study, in such a way that a fresh contribution to the body of knowledge is achievable. The point here is that the researcher must explore the literature broadly, while at the same time bearing in mind the constraints of the study.

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